Notes: Kehoe, Lopez, Midrigan & Pastorino (RES, 2023)

Asset Prices and Unemployment Fluctuations: A Resolution of the Unemployment Volatility Puzzle

Contents

1 Big picture

- **Question.** Can a search-and-matching model generate realistic unemployment volatility without relying on wage rigidity or inefficient bargaining, while also matching asset-pricing facts?
- **Puzzle.** The textbook Diamond–Mortensen–Pissarides model produces too little volatility in vacancies, job finding, and unemployment relative to productivity shocks. Existing fixes often conflict with the cyclicality of wages, the procyclicality of the opportunity cost of employment, or the volatility of risk-free rates.
- **Main answer.** The paper resolves the unemployment volatility puzzle by combining two ingredients: time-varying risk premia from asset-pricing preferences and human capital accumulation on the job.
- **Core mechanism.** Hiring a worker is like buying a risky long-duration asset. Because workers accumulate human capital while employed and lose it while unemployed, the surplus from a new match has long-duration payoffs. In recessions, the price of risk rises, so the value of these long-duration payoffs falls sharply. Firms therefore create fewer vacancies, job finding collapses, and unemployment rises.
- **No wage rigidity needed.** The paper uses competitive search, which delivers efficient wage setting. The mechanism does not require sticky wages, bargaining inefficiencies, or inefficient contracting.
- **Asset-pricing connection.** Preferences are chosen to generate time-varying risk premia, volatile stock returns, and stable risk-free rates. The same stochastic discount factor that prices assets also prices the long stream of match surplus flows.
- **Human capital connection.** Wages grow with labour market experience in the data. The model captures this by allowing human capital to grow during employment and depreciate during unemployment.
- **Main quantitative result.** The baseline model reproduces observed volatility in the job-finding rate and unemployment while matching key asset-price moments such as the equity premium, stock-return volatility, price-dividend ratio, and long-term bond yields.
- **Robustness.** The result survives extensions with physical capital, life-cycle workers, and alternative preference structures such as Campbell–Cochrane external habit, Epstein–Zin long-run risk, disaster risk, and affine stochastic discount factor specifications.
- **Takeaway.** Labour-market volatility and asset-price volatility are not separate puzzles. The paper shows that the same time-varying price of risk that helps explain asset prices can also generate large unemployment fluctuations once hiring is recognized as a long-duration risky investment.

2 Incremental contribution of the paper

2.1 Contribution relative to the Shimer puzzle literature

- Shimer (2005) showed that the canonical search model cannot produce enough unemployment volatility in response to plausible productivity shocks.
- Many proposed solutions increase amplification through wage rigidity, bargaining frictions, or changes in the opportunity cost of employment.
- Recent critiques show that many of these solutions are inconsistent with observed wage cyclicality, the procyclicality of the opportunity cost of employment, or the low volatility of risk-free rates.
- This paper offers a resolution that keeps wage setting efficient and keeps the opportunity cost of employment procyclical.

Interpretation: The incremental contribution is to solve the unemployment volatility puzzle without using the mechanisms that later evidence showed to be problematic. The paper restores the original promise of search theory: involuntary unemployment can fluctuate substantially in equilibrium without inefficient wage setting.

2.2 Contribution relative to macro-finance models

- Macro-finance models with habit, long-run risk, disasters, or affine stochastic discount factors explain asset-price volatility through time-varying risk premia.
- In standard real business cycle models, asset-pricing preferences often have little impact on real quantities, echoing the Tallarini separation result.
- This paper breaks that separation by embedding asset-pricing preferences in a search model with human-capital-driven long-duration match surplus.
- The stochastic discount factor does not only price financial assets; it also changes firms' valuation of hiring opportunities.

Interpretation: The incremental contribution is to create a real-financial interaction: asset-pricing forces affect unemployment because the job-creation decision is itself an asset-pricing decision.

2.3 Contribution relative to human capital and wage dynamics

- Existing labour-search models often treat workers as producing a constant flow after matching.
- The paper introduces human capital accumulation during employment and depreciation during unemployment, consistent with empirical wage-experience profiles.
- This human capital process gives match surplus flows long duration: early hiring creates payoff streams that extend far into the future.
- Long-duration surplus flows are highly sensitive to risk premia, which is why asset-pricing shocks translate into vacancy and unemployment fluctuations.

Interpretation: The paper's distinctive mechanism is not simply "risk premia matter." Risk premia matter because human capital makes the benefits of hiring long-lived.

2.4 Contribution relative to efficient wage-setting models

- The model uses competitive search, so allocations are efficient given the search technology.
- The authors emphasize that equivalent outcomes can arise under Nash bargaining or alternating-offer bargaining if those protocols satisfy the relevant efficiency conditions.
- Therefore the results are not tied to one special wage-determination protocol.

Interpretation: This is important because many unemployment-volatility explanations rely on wage rigidity or inefficient bargaining. Here, efficient wage setting still generates large labour-market volatility.

3 Data, calibration, and empirical targets

3.1 Main empirical moments

The baseline calibration targets three groups of moments:

- **Productivity moments:** mean productivity growth and standard deviation of productivity growth.

- **Labour-market moments:** mean unemployment, mean and volatility of job-finding rates, volatility of unemployment, autocorrelations, and the negative correlation between unemployment and job finding.
- **Asset-market moments:** equity premium, stock-return volatility, price-dividend ratio, long-term real and nominal yields, and their volatilities.

Interpretation: The model is disciplined by both real and financial data. It cannot solve unemployment volatility by generating counterfactual asset prices or risk-free-rate behaviour.

3.2 Key calibration choices

- Productivity follows a random walk with drift.
- The home-production parameter is fixed at $b = 0.6$, making the opportunity cost of employment procyclical with productivity.
- The matching-function elasticity is set to 0.5.
- The exogenous separation rate is set to about 2.8%.
- Human capital grows on the job at roughly 3.5% per year in the baseline calibration.
- The preference parameters are chosen to match asset-pricing moments and generate time-varying risk premia.

Interpretation: The model is not allowed to make unemployment volatile by making home production countercyclical or by making hiring costs unrelated to worker productivity. Both home production and vacancy costs scale with productivity and human capital.

3.3 Why asset-price data are central

- A high and volatile price of risk is directly disciplined by stock and bond return moments.
- In recessions, the stochastic discount factor places high value on current consumption and heavily discounts risky future payoffs.
- The model's labour-market mechanism depends on the same discounting of future match surplus.

Interpretation: Asset-price data are not an auxiliary check. They are central to pinning down the force that moves job creation.

4 Model setup and equations

4.1 Productivity process

Aggregate productivity follows:

$$\log(A_{t+1}) = g_a + \log(A_t) + \sigma_a \varepsilon^a_{t+1}, \qquad \varepsilon^a_{t+1} \sim N(0,1). \tag{1}$$

Interpretation: Productivity growth is the fundamental aggregate shock. The paper asks why plausible productivity shocks generate large unemployment movements once asset prices and human capital are incorporated.

4.2 Human capital accumulation and depreciation

A consumer has human capital z_t. If employed,

$$z_{t+1} = (1 + g_e)z_t. \tag{2}$$

If unemployed,

$$z_{t+1} = (1 + g_u)z_t, \tag{3}$$

where $g_e \geq 0$ and $g_u \leq 0$.

Interpretation: Employment raises future productivity and unemployment erodes it. This creates a wedge between the future surplus of current matches and the surplus from leaving workers unemployed.

4.3 Production, home production, and vacancies

An employed worker with human capital z produces:

$$A_t z. \tag{4}$$

An unemployed worker produces home output:

$$bA_t z, \qquad 0 < b < 1. \tag{5}$$

Posting a vacancy directed at a worker with human capital z costs:

$$\kappa A_t z. \tag{6}$$

Interpretation: Home production and hiring costs scale with aggregate productivity and human capital. This prevents the model from using countercyclical opportunity costs or artificially cheap recruiting to create amplification.

4.4 Employment and unemployment laws of motion

Let $u_t^b(z)$ be the measure of unemployed workers with human capital z at the beginning of period t:

$$u_t^b(z) = \frac{\phi}{1 + g_u} u_{t-1}\left(\frac{z}{1 + g_u}\right). \tag{7}$$

Employment evolves according to:

$$e_t(z) = \frac{\phi(1 - \sigma)}{1 + g_e} e_{t-1}\left(\frac{z}{1 + g_e}\right) + \lambda_t^w(\theta_t(z))u_t^b(z). \tag{8}$$

Unemployment evolves according to:

$$u_t(z) = \frac{\phi\sigma}{1 + g_e} e_{t-1}\left(\frac{z}{1 + g_e}\right) + [1 - \lambda_t^w(\theta_t(z))]u_t^b(z) + (1 - \phi)\nu(z). \tag{9}$$

Interpretation: Workers enter unemployment after separation, remain unemployed if they fail to match, and are replenished by new entrants. Human capital changes the distribution of workers across states.

4.5 Matching and market tightness

Market tightness for workers with human capital z is:

$$\theta_t(z) = \frac{v_t(z)}{u_t^b(z)}. \tag{10}$$

The job-finding rate is:

$$\lambda_t^w(\theta_t(z)) = \frac{m_t(u_t^b(z), v_t(z))}{u_t^b(z)}. \tag{11}$$

The job-filling rate is:

$$\lambda_t^f(\theta_t(z)) = \frac{m_t(u_t^b(z), v_t(z))}{v_t(z)}. \tag{12}$$

Interpretation: Competitive search lets firms direct vacancies to specific workers. Equilibrium tightness determines both workers' job-finding probabilities and firms' vacancy-filling probabilities.

4.6 Resource constraint

The aggregate resource constraint is:

$$C_t = A_t \int z e_t(z) dz + bA_t \int z u_t(z) dz - \kappa A_t \int z v_t(z) dz. \tag{13}$$

Interpretation: Output equals market production plus home production minus recruiting costs. Unemployment is costly because home production is less valuable than employment and because matching uses resources.

4.7 Preferences and stochastic discount factor

Utility is:

$$E_0 \sum_{t=0}^{\infty} \beta^t \frac{(C_t - X_t)^{1-\alpha}}{1-\alpha}. \tag{14}$$

Let S_t be the scaled surplus consumption ratio. Its law of motion is:

$$s_{t+1} = (1-\rho_s)s + \rho_s s_t + \lambda_a(s_t)[\Delta a_{t+1} - E_t(\Delta a_{t+1})], \tag{15}$$

where $s_t = \log S_t$ and $a_t = \log A_t$.

The stochastic discount factor is:

$$Q_{t,t+1} = \beta \left(\frac{S_{t+1}C_{t+1}}{S_t C_t} \right)^{-\alpha}. \tag{16}$$

Interpretation: The surplus consumption ratio makes risk premia time varying. In downturns, risk aversion rises and risky long-duration payoffs become much less valuable.

5 Equilibrium characterization and mechanism

5.1 Competitive search equilibrium

- Firms post wage offers and vacancies targeted at workers with different human-capital levels.
- Workers choose the labour markets that maximize their promised value.
- Free entry of vacancies and competitive wage offers determine efficient tightness.
- The equilibrium can be represented by an aggregate planning problem.

Interpretation: Competitive search is used because it delivers efficient wage setting. The model's unemployment volatility therefore does not arise from inefficient wage bargains.

5.2 Linearity and aggregation

Because output, home production, and vacancy costs scale linearly in human capital z, equilibrium values are linear in z.

Interpretation: This tractability result means the aggregate state can be summarized by the total human capital of employed workers and unemployed workers, rather than the full distribution of human capital.

5.3 Job-finding rate as an asset-pricing object

A key result is that the job-finding rate is approximately:

$$\log(\lambda_t^w) = \chi + \frac{1-\eta}{\eta} \log \left[\sum_{n=0}^{\infty} (c_\ell \delta_\ell^n + c_s \delta_s^n) \frac{P_t^n}{A_t} \right], \tag{17}$$

where P_t^n is the price of a claim to aggregate productivity n periods ahead.

Interpretation: Job creation is driven by the market value of future match surplus flows. The formula makes the job-finding rate an explicit weighted average of asset prices.

5.4 Why human capital creates duration

- Human capital accumulation makes a new worker more productive in future periods.
- Human capital depreciation during unemployment raises the cost of not employing a worker today.
- The surplus from a match therefore decays slowly over time.
- The job-finding rate places substantial weight on long-maturity productivity claims.

Interpretation: Duration is the bridge between asset pricing and labour markets. Long-duration surplus is highly sensitive to risk premia.

5.5 Price of productivity claims

The log-linear approximation for the price of an n-period productivity claim is:

$$\log\left(\frac{P_t^n}{A_t}\right) = a_n + b_n(s_t - s). \tag{18}$$

The elasticity b_n increases with maturity n.

Interpretation: Long-maturity productivity claims are much more sensitive to the surplus consumption state than short-maturity claims. When the state is low in a recession, long-duration match values fall sharply.

5.6 Sufficient statistic for job-finding volatility

The elasticity of the job-finding rate with respect to the surplus consumption state is:

$$\frac{d\log(\lambda_t^w)}{ds_t} = \frac{1-\eta}{\eta} \sum_{n=0}^{\infty} \omega_n b_n, \tag{19}$$

where the weights ω_n are proportional to the weighted prices of productivity claims.

The standard deviation of the job-finding rate is:

$$\sigma[\log(\lambda_t^w)] = \frac{d\log(\lambda_t^w)}{ds_t}\sigma(s_t). \tag{20}$$

Interpretation: A large response requires both large long-horizon price elasticities b_n and large long-horizon weights ω_n. Time-varying risk gives the first; human capital duration gives the second.

6 Quantitative findings and extensions

6.1 Baseline model

- The baseline model matches productivity growth, unemployment, job finding, and asset-price moments.
- It produces job-finding-rate volatility close to the data.
- It produces unemployment volatility close to the data.
- It matches the negative correlation between unemployment and job finding.

Interpretation: The baseline model solves the unemployment volatility puzzle while remaining disciplined by asset-price evidence.

6.2 Role of time-varying risk

- Replacing asset-pricing preferences with CRRA preferences removes the mechanism.
- Under CRRA, the job-finding rate and unemployment are nearly invariant to productivity shocks.
- The volatility of unemployment depends on the same risk forces that generate stock-market volatility and equity premia.

Interpretation: Time-varying risk is necessary. Human capital alone cannot produce large unemployment fluctuations under standard CRRA pricing.

6.3 Role of human capital acquisition

- Removing human capital accumulation eliminates almost all unemployment volatility.
- Human capital growth on the job and depreciation off the job generate duration in match surplus.
- Even modest rates of human capital accumulation and depreciation can generate large amplification.

Interpretation: Time-varying risk also is not enough by itself. Without human capital, match surplus is short duration, so changes in risk premia have little effect on hiring.

6.4 Physical capital extension

- The model is extended to include physical capital and investment adjustment costs.
- It matches consumption, investment, output, labour-market moments, stock returns, and bond yields.
- Physical capital does not undo the core mechanism.

Interpretation: The mechanism survives in a richer business-cycle environment with standard capital accumulation.

6.5 Life-cycle extension

- The life-cycle extension distinguishes young and mature workers.
- Young workers have higher human-capital growth and higher unemployment volatility.
- The model reproduces the empirical fact that unemployment volatility is higher for younger workers than for mature workers.

Interpretation: Human-capital dynamics explain not only aggregate unemployment volatility but also age-related differences in labour-market volatility.

6.6 Alternative preferences

- Campbell–Cochrane external habit produces results close to the baseline.
- Epstein–Zin long-run risk preferences also generate similar labour-market and asset-market moments.
- Disaster risk and affine stochastic discount factor specifications also preserve the mechanism.

Interpretation: The mechanism does not depend on one special preference specification. It requires time-varying risk premia and long-duration match surplus.

7 Tables and figures

7.1 Table 1: Baseline calibration and results

Read:

- Panel A reports parameter choices for the baseline model.
- The model targets productivity growth, productivity volatility, unemployment, the risk-free rate, excess returns, and price-dividend dynamics.
- The model matches labour-market moments closely: mean job-finding rate, volatility of job-finding, unemployment volatility, and the negative correlation between unemployment and job finding.
- The model also matches asset-market moments, including the equity premium and long-term bond yields.

Economic message: The baseline model simultaneously fits asset-price and labour-market facts. This is the paper's key quantitative achievement.

TABLE 1

Parametrization and results for baseline model

Panel A: Parameters		Panel B: Moments		
Endogenously chosen		*Targeted*	Data	Model
g_a, mean productivity growth (% p.a.)	2.28	Mean productivity growth (% p.a.)	2.28	2.28
σ_a, s.d. productivity growth (% p.a.)	1.84	S.d. productivity growth (% p.a.)	1.84	1.84
κ, hiring cost	1.00	Mean unemployment rate (%)	5.9	5.9
β, time preference factor	0.999	Mean risk-free rate (% p.a.)	0.74	0.74
S, mean of state S_t	0.21	S.d. risk-free rate (% p.a.)	2.35	2.35
α, inverse EIS	5.35	Sharpe ratio of stock market excess return (p.a.)	0.45	0.45
ρ_s^{12}, annualized persistence of state	0.935	Autocorrelation log price-dividend ratio (p.a.)	0.95	0.95
Assigned		*Labor market results*		
B, matching function efficiency	0.46	Mean job-finding rate (%)	45.35	50.14
b, home production parameter	0.6	S.d. job-finding rate (%)	6.67	6.60
σ, probability of separation	0.028	Autocorrelation job-finding rate	0.94	0.99
η, matching function elasticity	0.5	S.d. unemployment rate (%)	0.79	0.75
ϕ, survival probability	0.9972	Autocorrelation unemployment rate	0.96	0.99
g_e, human capital growth on job (% p.a.)	3.5	Correlation unemployment and job-finding rate	-0.96	-0.99
		Elasticity user cost of labor to unemp. (Basu and House, 2016)	-5.80	-6.00
		Asset market results		
		Mean excess return (% p.a.)	7.05	6.15
		S.d. excess return (% p.a.)	15.6	13.7
		Mean log price-dividend ratio	3.51	3.42
		S.d. log price-dividend ratio	0.44	0.38
		Mean 20-year real yield (% p.a.)	4.74	3.55
		S.d. 20-year real yield (% p.a.)	1.95	2.24
		Mean 20-year nominal yield (% p.a.)	7.71	7.53
		S.d. 20-year nominal yield (% p.a.)	2.41	2.30

We interpret each model year as corresponding to one year of (potential) labor market experience—measured in the data as age minus education minus six—and choose the survival probability ϕ to be consistent with an average working life of 30 years, which corresponds to the prime-age working horizon between the ages of 25 and 55 years. Below, we consider an alternative parametrization with ϕ set at a value consistent with an average working life of 40 years, as we assume in our life-cycle extension of the baseline model. We select a growth rate of human capital during employment g_e of 3.5% per year, which, taking into account an aggregate

v

7.2 Figure 1 and Table 2: time-varying risk and human capital

Read:

- Figure 1 shows that reducing the price of risk or reducing surplus-consumption volatility lowers both stock-market volatility and job-finding volatility.
- This links labour-market volatility directly to asset-market volatility.
- Table 2 shows that without human capital accumulation, the DMP model with baseline preferences produces almost no job-finding or unemployment volatility.
- The baseline model with human capital reproduces the key labour-market moments.

Economic message: Both time-varying risk and human capital are necessary. Risk premia create sensitivity to state variables; human capital makes hiring a long-duration investment.

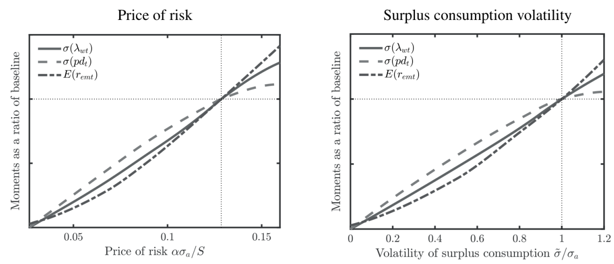

FIGURE 1

Sensitivity of key moments to preference parameters in baseline model

Note: λ_{wt} denotes the job-finding rate, pd_t denotes the log price-dividend ratio of the consumption claim,

Figure 1: relation between labour and stock-market volatility

TABLE 2

Role of human capital accumulation

	Data	Baseline	DMP model with baseline preferences ($g_e = 0$ and $g_u = 0$)	Baseline model with $g_e = 3.5\%$ and $g_u = 3.5\%$
S.d. job-finding rate (%)	6.67	6.60	0.14	0.14
Autocorrelation job-finding rate	0.94	0.99	0.99	0.99
S.d. unemployment rate (%)	0.79	0.75	0.02	0.02
Autocorrelation unemployment rate	0.96	0.99	0.99	0.99
Correlation unemployment and job-finding rate	-0.96	-0.99	-1.00	-1.00

Table 2: role of human capital accumulation

7.3 Figures 2 and 3: impulse responses and human-capital loci

Read:

- Figure 2 compares the baseline model with a DMP model after a negative productivity shock.
- The baseline model produces a large decline in the job-finding rate and a large rise in unemployment.
- The DMP version produces almost no labour-market response.
- Figure 3 maps combinations of human capital growth on the job and depreciation off the job that produce various fractions of baseline job-finding volatility.
- Even modest human capital dynamics can generate substantial amplification.

Economic message: Human capital dynamics are not a cosmetic addition. They are the quantitative source of long-duration surplus and labour-market amplification.

apital process than in the absence of it.

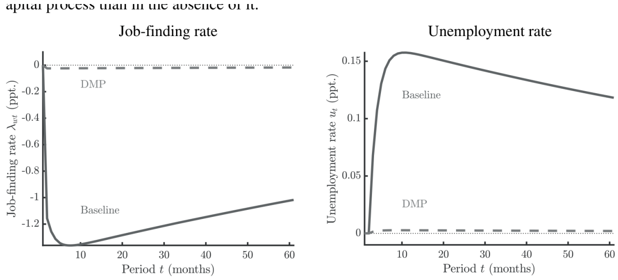

FIGURE 2

Labor market responses to productivity shock in baseline model and DMP model with baseline preferences

Figure 2: labour-market impulse responses

discussed, we adjust to keep the mean unemployment rate unchanged.

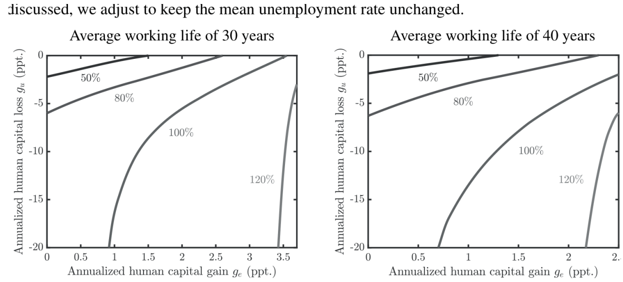

FIGURE 3

Loci of human capital rates corresponding to fractions of job-finding rate volatility in baseline model

Figure 3: loci of human-capital rates

7.4 Figures 4 and 5: asset prices of productivity claims

Read:

- Figure 4 shows that prices of longer-maturity productivity claims are much more sensitive to the surplus-consumption state than short-maturity claims.
- The approximate solution closely tracks the global solution.
- Figure 5 shows that long-horizon productivity-claim prices fall sharply after a negative productivity shock.
- The baseline model places meaningful cumulative weight on long maturities, whereas the DMP model places almost all weight on short maturities.

Economic message: The mechanism operates through duration. Long-maturity claims fall in value in recessions, and human capital makes job creation depend on those long-maturity claims.

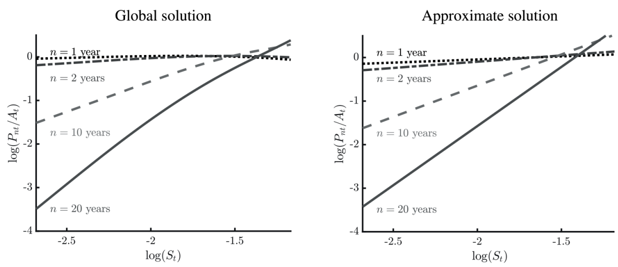

FIGURE 4

Figure 4: prices of productivity claims

elasticities of the prices of claims to future productivity, and so the weights on the prices of claims in the surplus flows in (3.38), are sufficiently large for large n.

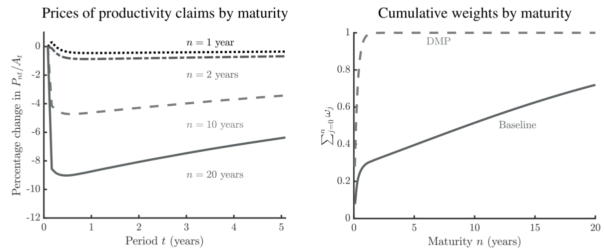

Figure 5: asset-market responses and duration weights

7.5 Table 3: Model with physical capital

Read:

- The physical-capital version matches productivity, unemployment, labour-market volatility, investment volatility, and asset-price moments.
- It produces job-finding-rate volatility of about 7.24%, close to the data.
- It produces unemployment volatility of about 0.84%, close to the data.
- It matches the equity premium and long-term nominal yields reasonably well.

Economic message: The core mechanism is robust to standard capital accumulation and investment adjustment costs.

Parametrization and results for model with baseline preferences and physical capital

Panel A: Parameters		Panel B: Moments		
Endogenously chosen		*Targeted*	Data	Model
g_a, mean productivity growth (% p.a.)	1.36	Mean productivity growth (% p.a.)	1.36	1.36
σ_a, s.d. productivity growth (% p.a.)	1.79	S.d. productivity growth (% p.a.)	1.79	1.79
κ, hiring cost	1.74	Mean unemployment rate (%)	5.9	5.9
β, time preference factor	0.999	Mean risk-free rate (% p.a.)	0.74	0.74
S, mean of state S_t	0.29	S.d. risk-free rate (% p.a.)	2.35	2.35
α, inverse EIS	7.15	Sharpe ratio of stock market excess return (p.a.)	0.45	0.45
ϑ, curvature of production function	0.26	Mean labor share of output	0.70	0.70
ξ, curvature of capital adjustment cost	0.26	Ratio s.d. investment to consumption growth	4.5	4.5
ρ_s^{12}, annualized persistence of state	0.935	Autocorrelation log price-dividend ratio (p.a.)	0.95	0.95
Assigned		*Labor market results*		
B, matching function efficiency	0.46	Mean job-finding rate (%)	45.35	49.85
b, home production parameter	0.6	S.d. job-finding rate (%)	6.67	7.24
σ, probability of separation	0.028	Autocorrelation job-finding rate	0.94	0.99
η, matching function elasticity	0.5	S.d. unemployment rate (%)	0.79	0.84
ϕ, survival probability	0.9972	Autocorrelation unemployment rate	0.96	0.99
g_e, human capital growth on job (% p.a.)	3.5	Correlation unemployment and job-finding rate	-0.96	-0.98
δ, physical capital depreciation rate	0.1/12			
		Asset market results		
		Mean excess return (% p.a.)	7.05	5.52
		S.d. excess return (% p.a.)	15.6	12.2
		Mean log price-dividend ratio	3.51	3.27
		S.d. log price-dividend ratio	0.44	0.36
		Mean 20-year real yield (% p.a.)	4.74	3.68
		S.d. 20-year real yield (% p.a.)	1.95	2.25
		Mean 20-year nominal yield (% p.a.)	7.71	7.66
		S.d. 20-year nominal yield (% p.a.)	2.41	2.30

As shown in Table 3, this augmented model gives rise to a volatility of both the job-finding

TABLE 4

Parametrization and results for life-cycle model with baseline preferences

Panel A: Parameters		Panel B: Moments		
Endogenously chosen		*Targeted*	Data	Model
g_a, mean productivity growth (% p.a.)	1.95	Mean productivity growth (% p.a.)	1.95	1.95
σ_a, s.d. productivity growth (% p.a.)	1.40	S.d. productivity growth (% p.a.)	1.40	1.40
κ_y, hiring cost for young	1.42	Mean unemployment rate for young (%)	8.7	8.7
κ_m, hiring cost for mature	6.01	Mean unemployment rate for mature (%)	4.1	4.1
β, time preference factor	0.998	Mean risk-free rate (% p.a.)	1.53	1.53
S, mean of state S_t	0.12	S.d. risk-free rate (% p.a.)	1.84	1.84
α, inverse EIS	4.00	Sharpe ratio of stock market excess return (p.a.)	0.45	0.45
ρ_s^{12}, annualized persistence of state	0.935	Autocorrelation log price-dividend ratio (p.a.)	0.96	0.96
σ_y, probability of separation for young	0.044	Mean job-finding rate for young (%)	46.87	46.61
σ_m, probability of separation for mature	0.015	Mean job-finding rate for mature (%)	34.95	35.55
Assigned		*Labor market results*		
B, matching function efficiency	0.42	S.d. job-finding rate for young (%)	4.98	5.46
b, home production parameter	0.6	S.d. job-finding rate for mature (%)	4.60	4.43
η, matching function elasticity	0.5	Autocorr. job-finding rate for young	0.94	0.99
ϕ_y, survival probability for young	0.9917	Autocorr. job-finding rate for mature	0.89	0.99
ϕ_m, survival probability for mature	0.9972	S.d. unemployment rate for young (%)	0.86	0.97
g_{ey}, HK growth on job for young (% p.a.)	4.22	S.d. unemployment rate for mature (%)	0.51	0.52
g_{uy}, HK growth off job for mature (% p.a.)	-0.99	Autocorr. unemployment rate for young	0.95	0.99
g_{em}, HK growth on job for young (% p.a.)	2.75	Autocorr. unemployment rate for mature	0.95	0.99
g_{um}, HK growth off job for mature (% p.a.)	-18.2	Corr. unemployment and job-finding rate for young	-0.95	-0.99
		Corr. unemployment and job-finding rate for mature	-0.93	-0.99
		Asset market results		
		Mean excess return (% p.a.)	5.94	4.27
		S.d. excess return (% p.a.)	13.2	9.50
		Mean log price-dividend ratio	3.70	3.45
		S.d. log price-dividend ratio	0.49	0.30
		Mean 20-year real yield (% p.a.)	4.74	3.17
		S.d. 20-year real yield (% p.a.)	1.95	1.37
		Mean 20-year nominal yield (% p.a.)	7.71	7.45
		S.d. 20-year nominal yield (% p.a.)	2.41	1.46

In this simple exercise, we have examined whether our basic mechanism can account for

7.6 Table 4: Life-cycle model

Read:

- The life-cycle model distinguishes young and mature workers.
- It matches higher unemployment and job-finding volatility for young workers relative to mature workers.
- It also preserves the asset-market implications of the baseline model.

Economic message: Human-capital dynamics naturally explain why labour-market volatility differs by age and experience.

7.7 Table 5: Alternative preference structures

Read:

- The model remains successful under Campbell–Cochrane external habit preferences.
- It also works under Epstein–Zin long-run risk, disaster risk, and affine SDF specifications.
- Labour-market volatility and asset-market moments are broadly similar across preference structures.

Economic message: The result is not tied to one specific preference model. The general requirement is a time-varying price of risk.

Results for other preferences

	Data	Baseline	Alternative preferences			
			CC	EZ w/ LRR	EZ w/ disasters	Affine SDF
Labor market results						
S.d. job-finding rate (%)	6.67	6.60	6.66	6.60	5.66	7.08
Autocorrelation job-finding rate	0.94	0.99	0.99	0.99	0.99	0.99
S.d. unemployment rate (%)	0.79	0.75	0.77	0.70	0.78	0.72
Autocorrelation unemployment rate	0.96	0.99	0.99	0.99	0.99	0.99
Correlation unemployment and job-finding rate	-0.96	-0.99	-0.98	-0.98	-0.98	-0.98
Asset market results						
Mean excess return (% p.a.)	7.05	6.15	6.32	4.26	4.77	6.92
S.d. excess return (% p.a.)	15.6	13.7	14.1	9.50	10.7	15.6
Mean log price-dividend ratio	3.51	3.42	3.41	3.85	3.76	3.29
S.d. log price-dividend ratio	0.44	0.38	0.39	0.31	0.36	0.38
Mean 20-year real yield (% p.a.)	4.74	3.55	3.70	2.79	-1.62	4.10
S.d. 20-year real yield (% p.a.)	1.95	2.24	2.40	1.26	2.15	2.07
Mean 20-year nominal yield (% p.a.)	7.71	7.53	7.67	6.48	2.04	8.19
S.d. 20-year nominal yield (% p.a.)	2.41	2.30	2.47	1.27	2.16	2.20

8 Short summary

- The paper resolves the unemployment volatility puzzle by linking labour-market fluctuations to asset-pricing forces.
- It adds time-varying risk premia and human capital acquisition to a competitive-search labour market.
- Hiring is a risky long-duration investment because human capital grows on the job and depreciates off the job.
- In recessions, the price of risk rises and the value of long-duration match surplus falls sharply.
- Firms respond by posting fewer vacancies, reducing job finding and raising unemployment.
- The baseline model matches job-finding volatility, unemployment volatility, and key asset-price moments.
- Removing time-varying risk or removing human capital destroys the result.
- The mechanism survives physical capital, life-cycle heterogeneity, and alternative preference structures.
- The paper's main lesson is that unemployment volatility and asset-price volatility can have a common source: variation in the price of risk.

9 Conclusion

9.1 Core conclusion

Kehoe, Lopez, Midrigan, and Pastorino show that the unemployment volatility puzzle can be resolved without wage rigidity or inefficient bargaining. Once hiring is treated as a risky long-duration investment, time-varying risk premia can produce large cyclical changes in vacancies, job finding, and unemployment.

9.2 Economic intuition

Human capital accumulation makes a newly created job valuable far into the future. Asset-pricing preferences imply that future risky payoffs become much less valuable in recessions. Firms therefore cut job creation sharply when the price of risk rises. This amplifies productivity shocks into large labour-market fluctuations.

9.3 Incremental contribution

The paper's incremental contribution is to unify two previously separate literatures. From macro-labour economics, it takes the Shimer unemployment volatility puzzle and the requirement that wages and opportunity costs behave realistically. From macro-finance, it takes time-varying risk premia that match asset prices. Its key new insight is that human capital gives match surplus long duration, allowing asset-pricing forces to move labour-market quantities in an efficient-search model.

9.4 Takeaway

The paper changes how to think about the unemployment volatility puzzle. The issue is not only wage determination or matching technology. It is also the valuation of long-duration labour-market investments under time-varying risk.